Biofilm and Oral Flora: Normal vs Pathologic Flora

Integrated Biomedical and Clinical Sciences 2 OMFS 513

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Learning Objectives and Competency Statements

Learning Outcomes

1. Describe different oral microbiota and their roles in oral health and diseases.
2. Describe main characters of oral microorganisms and their importance in dentistry.

Competency statements

Define and integrate the knowledge of microflora and their impact on oral health and disease.

Oral Microbiota

-Oral cavity contains many bacterial species:

Most are normal flora; a few are pathogenic (associated with dental and periodontal diseases)

-Commensal (non-pathogenic) bacteria:

Compete with harmful microbes for adhesion sites

Produce toxic metabolites that inhibit pathogen growth

Provide protective functions

-Dental caries and periodontal disease:

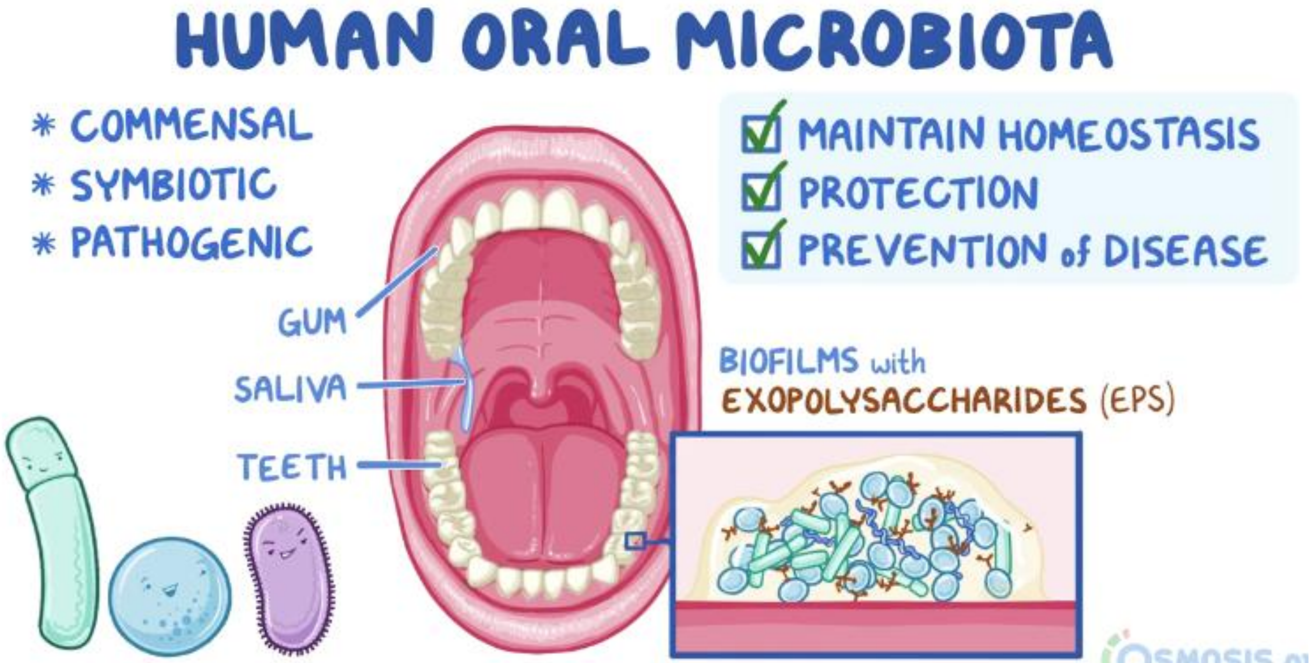
Among the most common infections

Significant economic burden

-Dental plaque:

Diverse bacteria adhere to teeth

Form a complex biofilm



Factors Influencing Oral Microbiota

-Oral environment changes frequently:

Temperature, oxygen levels, pH
Puberty: Hormones
Immune and non-immune defense factors
Diet
Salivary flow rate

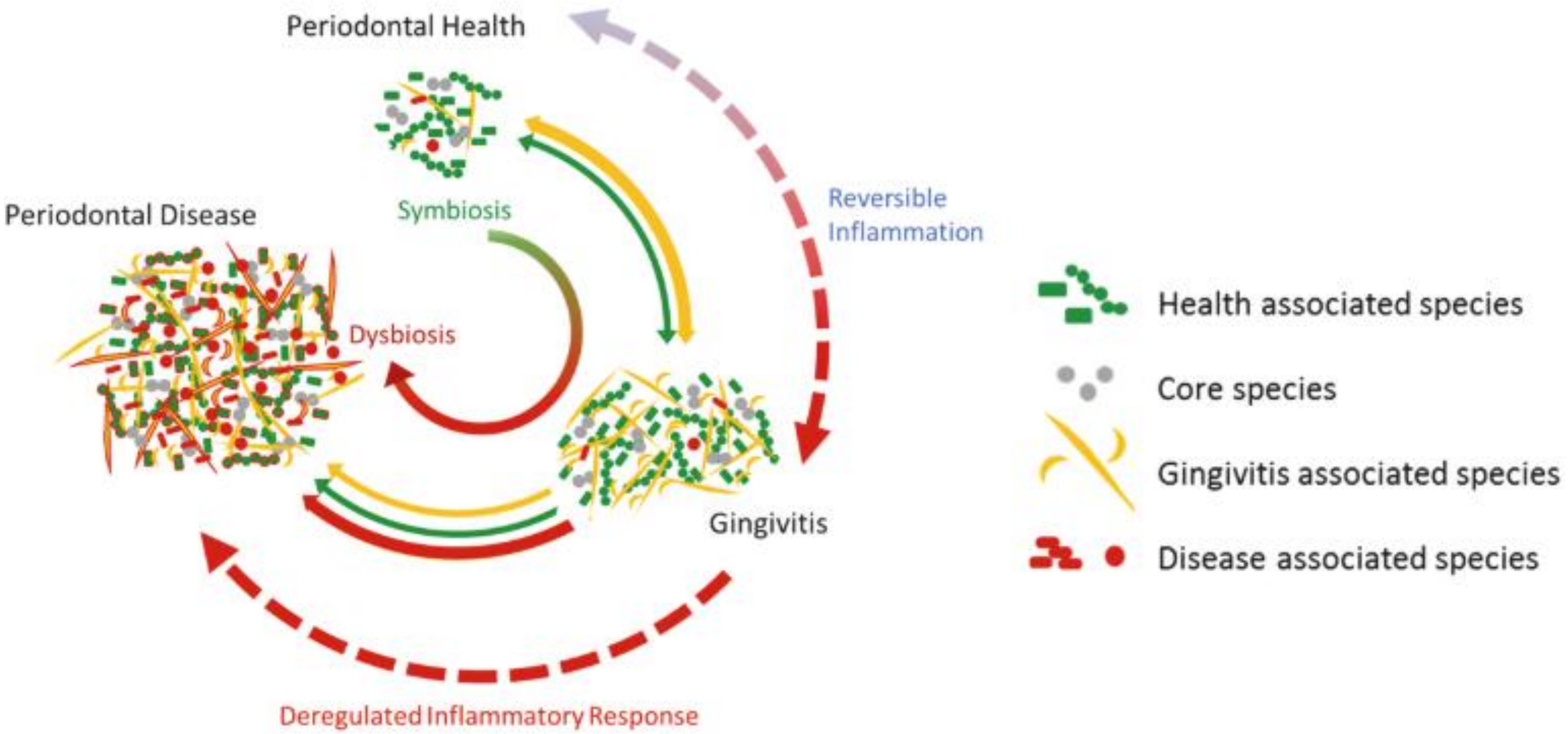
-Colonization timeline:

At birth: minimal bacterial presence
After birth: continuous bacterial introduction
Primary teeth allow stable colonization

-Microbiota shifts across lifespan:

Influenced by age, physical activity, stress
Lifestyle factors: smoking, frequent sugar intake, pregnancy

Image is for illustrative purposes



-Most organisms survive in oropharynx by **adhering to soft and hard tissues**.

-Without adhesion, they are cleared by **swallowing, chewing, nose blowing, oral hygiene**, saliva flow, crevicular fluid, and **cilia movement**.

-Oral bacteria, including pathogens like *P. gingivalis* and *A. actinomycetemcomitans*, can **adhere to intraoral surfaces** and **mucosa**.

-Oral bacteria increase with exposure to environmental microbes.

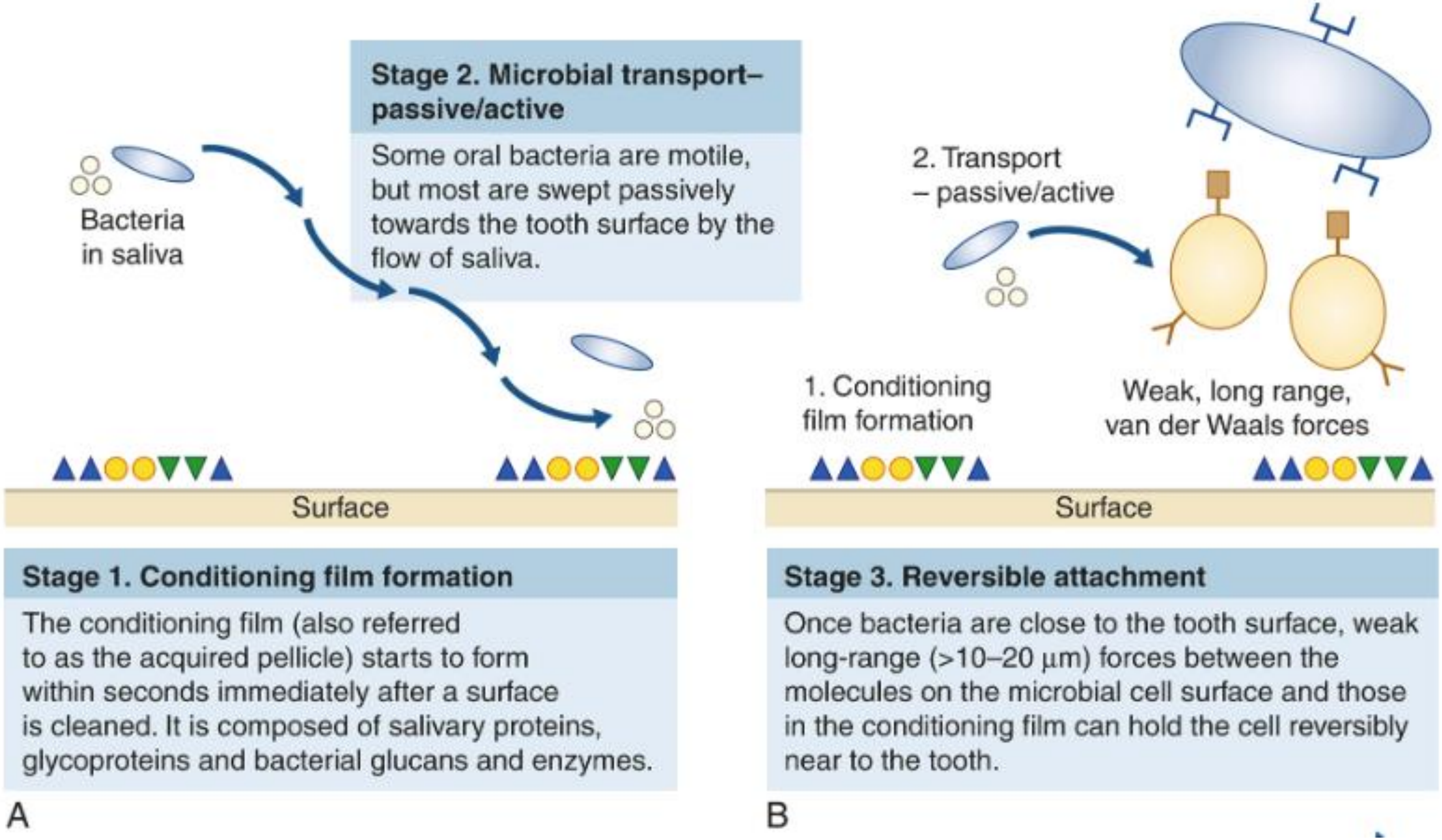
-Early colonizers: *Streptococcus salivarius*, *S. mitis*, *Veillonella*, *Neisseria*, *Actinomyces*, *Staphylococcus*.

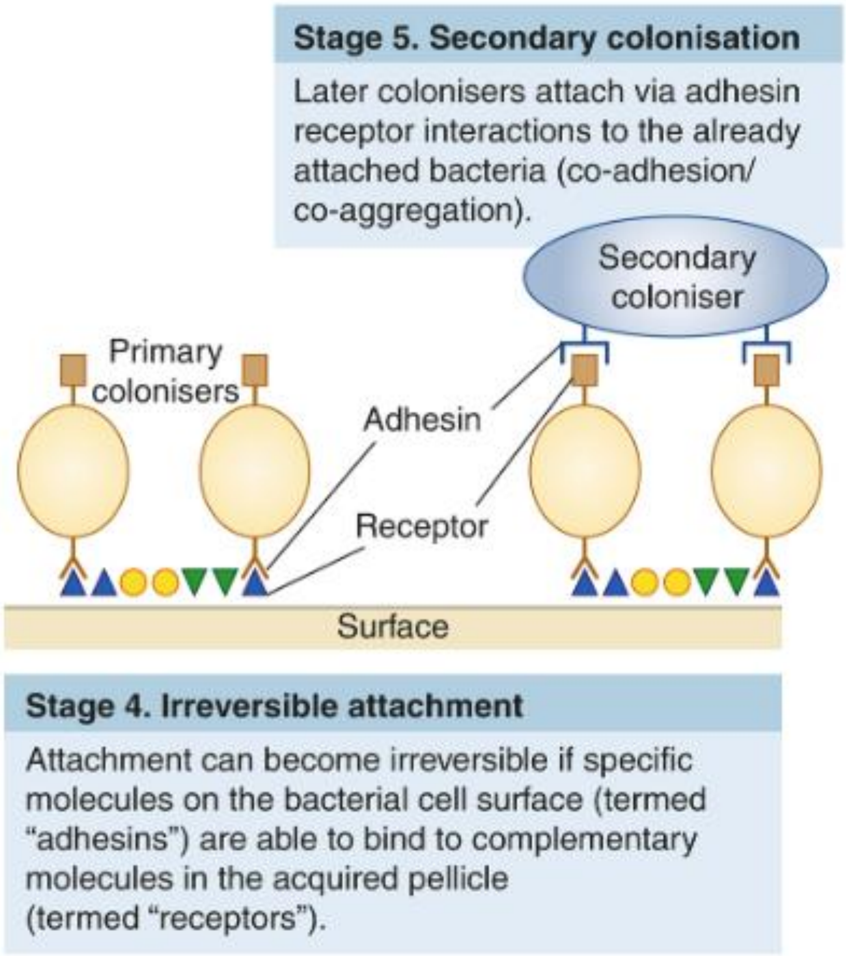
-After tooth eruption: more complex microbiota develops, including *S. sanguinis*, *Lactobacillus*, *S. oralis*.

-After first year of life: oral streptococci such as *S. oralis*, *S. anginosus*, *S. mutans*, *S. sobrinus*, *S. gordonii* become common.

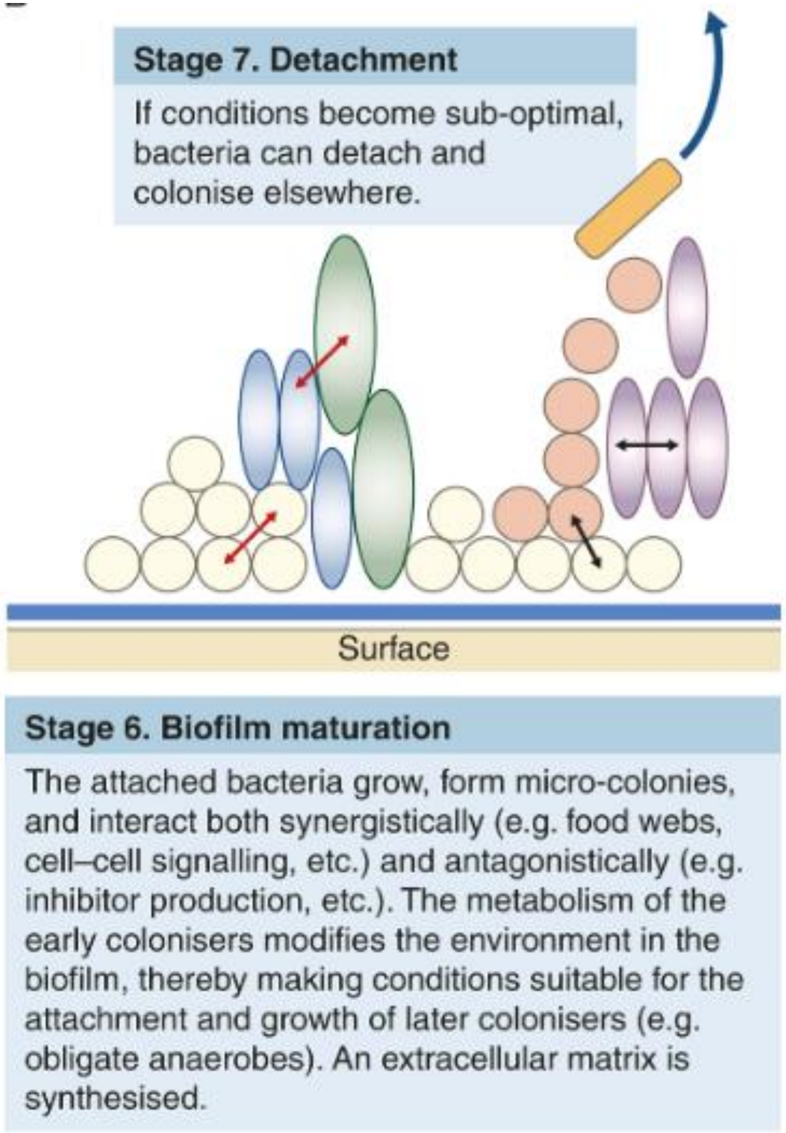
-Anaerobes: like *Fusobacterium* and *Prevotella* appear; diversity increases with more erupted teeth.

-Adult oral microbiome: ~700 species, mostly harmless commensals. Imbalance (e.g., after antibiotics) can allow pathogens or Candida to overgrow.


Stage 2. Microbial transport–passive/active
Some oral bacteria are motile, but most are swept passively towards the tooth surface by the flow of saliva.
Bacteria in saliva
Surface
Stage 1. Conditioning film formation
The conditioning film (also referred to as the acquired pellicle) starts to form within seconds immediately after a surface is cleaned. It is composed of salivary proteins, glycoproteins and bacterial glucans and enzymes.
A
2. Transport – passive/active
1. Conditioning film formation
Weak, long range, van der Waals forces
Surface
Stage 3. Reversible attachment
Once bacteria are close to the tooth surface, weak long-range (>10–20 µm) forces between the molecules on the microbial cell surface and those in the conditioning film can hold the cell reversibly near to the tooth.
B


Stage 5. Secondary colonisation
Later colonisers attach via adhesin receptor interactions to the already attached bacteria (co-adhesion/ co-aggregation).
Secondary coloniser
Primary colonisers
Adhesin
Receptor
Surface
Stage 4. Irreversible attachment
Attachment can become irreversible if specific molecules on the bacterial cell surface (termed "adhesins") are able to bind to complementary molecules in the acquired pellicle (termed "receptors").

C


Stage 7. Detachment
If conditions become sub-optimal, bacteria can detach and colonise elsewhere.
Surface
Stage 6. Biofilm maturation
The attached bacteria grow, form micro-colonies, and interact both synergistically (e.g. food webs, cell–cell signalling, etc.) and antagonistically (e.g. inhibitor production, etc.). The metabolism of the early colonisers modifies the environment in the biofilm, thereby making conditions suitable for the attachment and growth of later colonisers (e.g. obligate anaerobes). An extracellular matrix is synthesised.

D

Oral Cavity Ecosystems (Niches)

*Oral cavity divided into **6 major ecosystems (niches)**, with distinctive ecological determinants:

- Intraoral and supragingival hard surfaces (teeth, Implants, restorations, prostheses)
- Subgingival region adjacent to a hard surface: periodontal/peri-Implant pockets (presence of crevicular fluid, root cementum or Implant surface, pocket epithelium)
- Buccal palatal epithelium and floor of the mouth epithelium
- Dorsum of the tongue
- Tonsils
- Saliva

-Teeth and Implants:

Provide hard, non-shedding surfaces for bacterial buildup

Create a unique break in the ectodermal barrier

Bacterial accumulation on these surfaces drives **caries, gingivitis, periodontitis, peri-implantitis**, and **halitosis**

-Soft tissues:

Support bacterial adhesion and early colonization

Natural cleansing occurs through **epithelial shedding** (twice daily)

-Areas with reduced cleansing:

Tongue surface and spaces under removable prostheses trap shed cells

Brushing these areas is recommended

-Healthy oral microbiome includes major phyla: Firmicutes, Proteobacteria, Actinobacteria, Bacteroidetes, Fusobacteria

-Periodontal health: dominated by **gram-positive facultative bacteria** (Streptococcus, Actinomyces) with smaller amounts of **gram-negative species**

-Protective species: *S. sanguinis*, *C. ochracea*, *Veillonella parvula* may help prevent colonization by pathogens

FIGURE 7.3.4 Peri-implant health in an edentulous mandible in a patient with a number of risk factors, including xerostomia, but with excellent oral hygiene. This patient had been provided with a mandibular bar and clip-retained over-denture to allow for proper self-care and professional maintenance.

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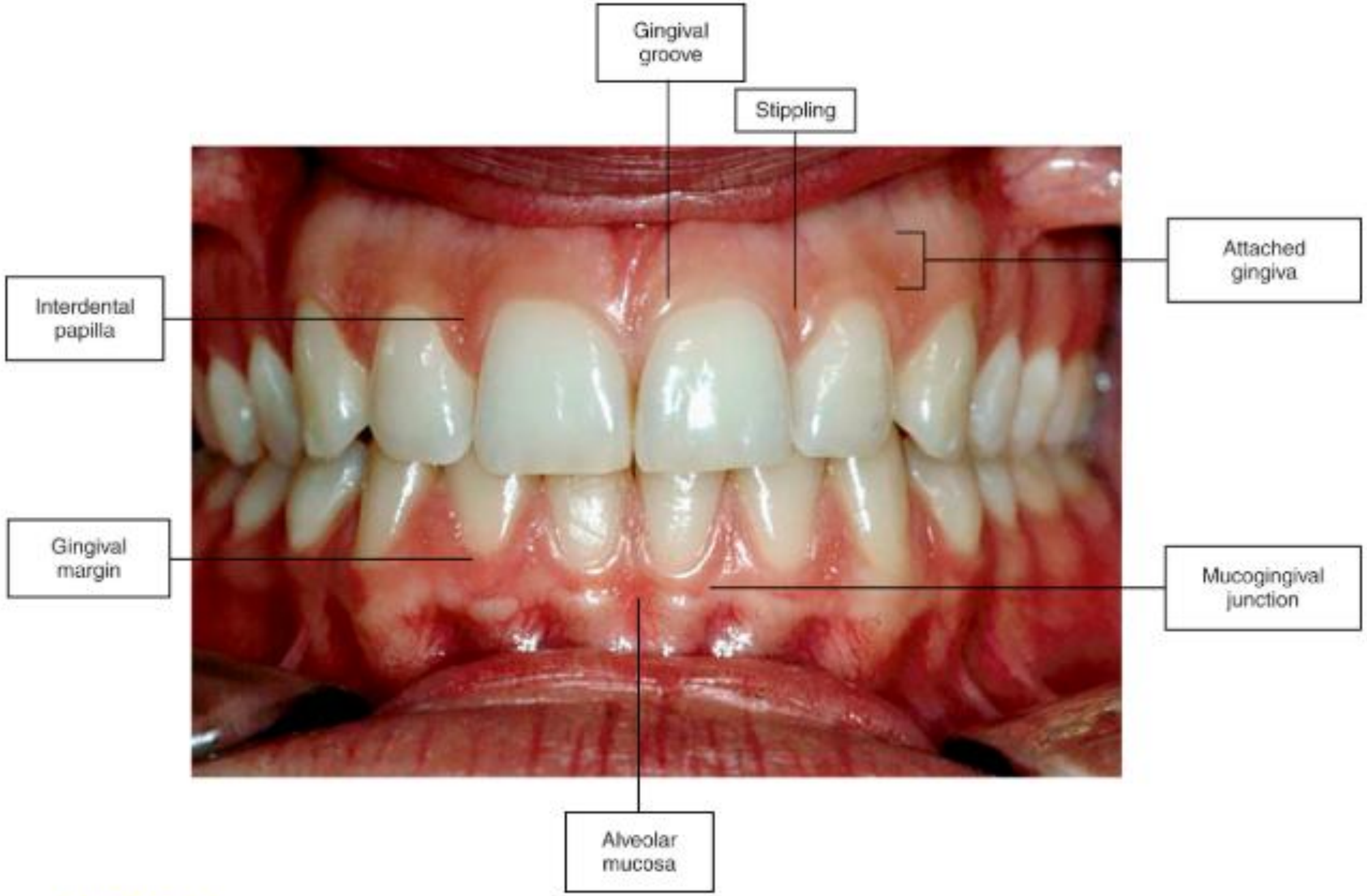


FIGURE 1.1.3 Healthy gingival tissues (Reproduced with kind permission by Quintessence from Chapple, I.L.C., Gilbert A.D., 2002. *Understanding Periodontal Diseases: Assessment and Diagnostic Procedures in Dental Practice*. Quintessence, London. ISBN: 1-85097-053-X).

Pellicle, Materia Alba, Biofilm

-Pellicle:

Thin organic layer forms on all oral surfaces within 1 minute

Made of peptides, proteins, glycoproteins

Functions: prevents drying, lubricates, promotes bacterial adhesion

-Materia alba:

Soft, loosely attached mix of bacteria, food debris, shed cells

Not organized like plaque and easily rinsed away

-Biofilm:

Microbial community embedded in a protective matrix (polysaccharides, proteins, nucleic acids)

Bacteria interact, communicate, and exchange DNA

Matrix retains and concentrates bacterial products

Plaque

-Dental plaque:

Structured, resilient deposit on hard surfaces
Cannot be removed by rinsing
Primary cause of **caries** and **periodontal diseases**
Typical **biofilm** containing bacteria and their products (e.g., glucans)

-Intercellular matrix:

Organic: polysaccharides (dextran), proteins, glycoproteins, lipids, DNA
Inorganic: calcium, phosphorus, sodium, potassium, fluoride
Derived from saliva, GCF, and bacterial products

-Plaque classification:

Supragingival: at/above gingival margin
Subgingival: below margin, within pocket

-Microbial differences:

Subgingival plaque has **more anaerobic species** due to blood products and low oxygen

Accumulation plaque biofilm

-Plaque and caries:

Plaque contains bacteria + acids from fermentable carbs
Acids lower enamel pH → decalcification/demineralization and caries
Fluoride, calcium, phosphate in saliva support remineralization

-Initial bacterial adhesion/attachment:

Bacteria attach to the pellicle; early colonizers are mainly Streptococcus spp.
Early plaque is dominated by obligate aerobes and facultative anaerobes
These bacteria change the environment (e.g., reduce oxygen), allowing anaerobes to thrive later
Early colonizers use sugars; mature plaque bacteria are obligate anaerobic and asaccharolytic (use amino acids/peptides)

-Colonization and maturation:

First colonizers: **Gram-positive facultative species**: Streptococcus, Actinomyces, Lactobacillus (bind directly to pellicle)

Secondary colonizers: attach to existing bacteria, not clean tooth surfaces

-Bacterial colonization: begins within hours after pellicle forms; plaque matures as additional species multiply

-Microbial shift over time: early **gram-positive facultative anaerobic** bacteria decrease; **gram-negative anaerobes** increase—especially at gingival margins

-Plaque is **dynamic:** shifts in microbial composition contribute to oral diseases

-Survival traits: successful colonizers must **adhere** and adapt to oral surfaces

-Caries and **periodontal disease:** both are **polymicrobial** infections

-Plaque maturation: early supragingival plaque → mature subgingival plaque with a shift from gram-positive to gram-negative species

-Calculus formation: mineralized plaque becomes **calculus**; covered by unmineralized plaque

-Subgingival calculus often **dark green/brown** due to blood products associated with **subgingival hemorrhage**

Calculus

-Plaque mineralization:

~50% mineralized in 2 days; ~90% in 12 days

-Supragingival calculus:

White/yellow, clay-like

Above gingival margin

Removed by professional cleaning

Common sites: buccal of maxillary molars and lingual of mandibular incisors (near major salivary ducts)

-Subgingival calculus:

Hard, dense, dark brown/green-black

Covered by unmineralized plaque → rough surface for pathogenic bacteria

Not removable by rinsing

Removal reduces inflammation and probing depths

-Composition:

70–90% inorganic: calcium phosphate (76%), calcium carbonate, magnesium phosphate, other minerals

10–30% organic: protein-polysaccharide complexes, epithelial cells, leukocytes, microorganisms

-Mineral sources:

Supragingival: mainly saliva

Subgingival: gingival crevicular fluid

Health

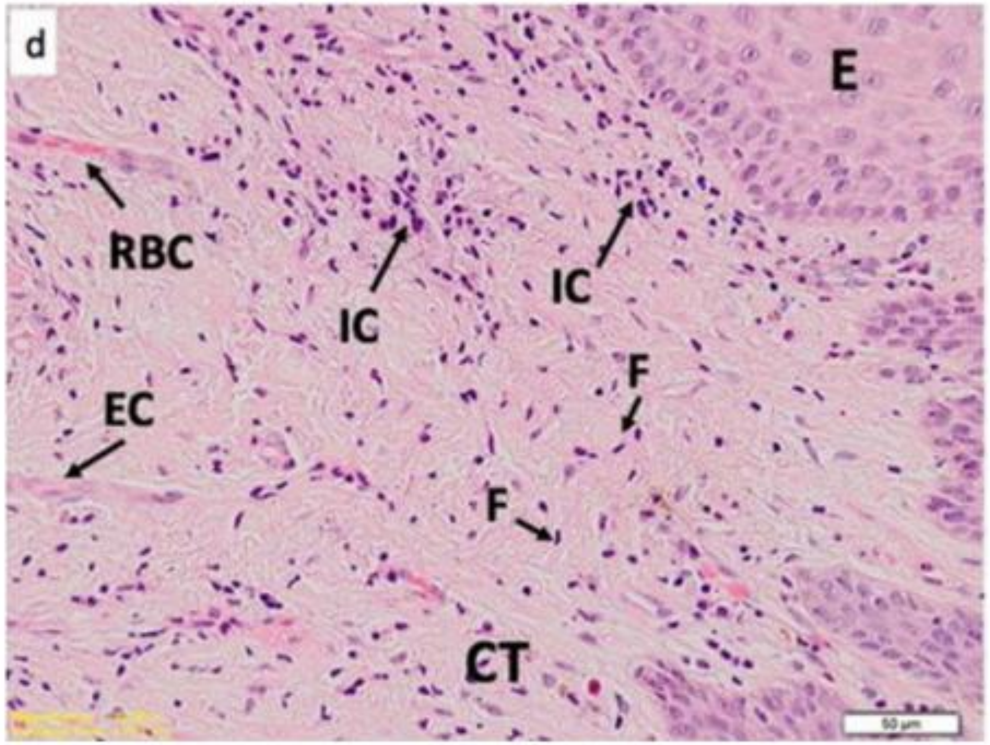
Courtesy of Dr. Wu

-No bleeding, inflammation, or loss of bone
-When you see a patient and begin to probe their mouth, this is what you will see if they are periodontally healthy (no bleeding on probing and probing depth ranges from 1-3 mm)

Composition of healthy gingiva

- Predominantly T cells
- Few B cells or plasma cells

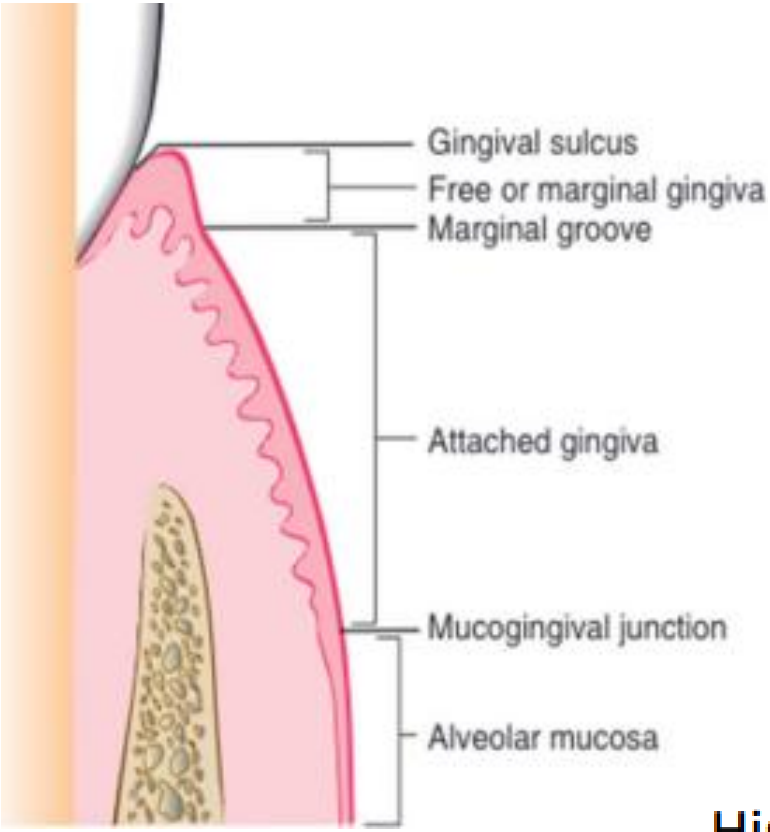
Role: Plays a part in daily host response to bacteria



Lee, Y.H., Baharuddin, N.A., Chan, S.W. et al. Localisation of citrullinated and carbamylated proteins in inflamed gingival tissues from rheumatoid arthritis patients. Clin Oral Invest 25, 1441–1450 (2021).

How do you measure periodontal health?

Courtesy of Dr. Wu



-Gingival sulcus is a natural space between soft tissue and area surrounding tooth

-Clinically, you probe the depth of gingival sulcus to determine periodontal health

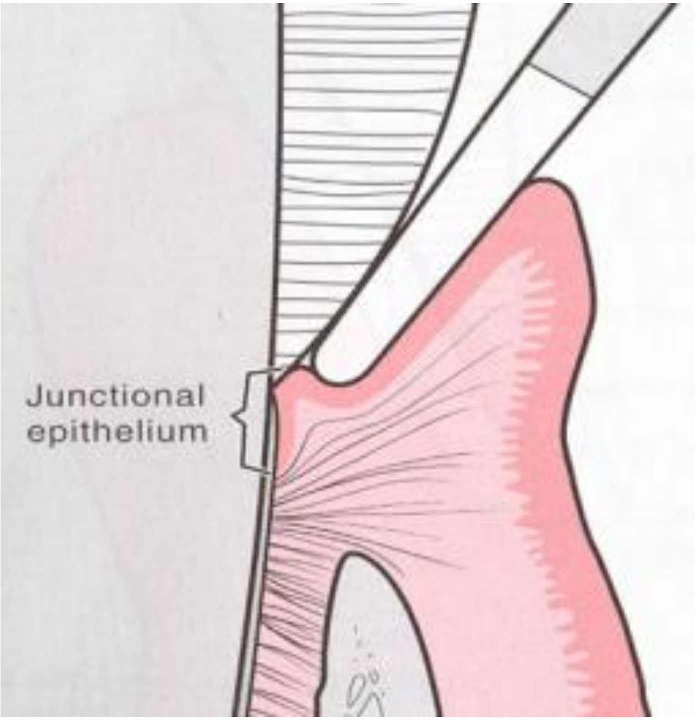
-Probing depth is an important diagnostic tool

Histological:

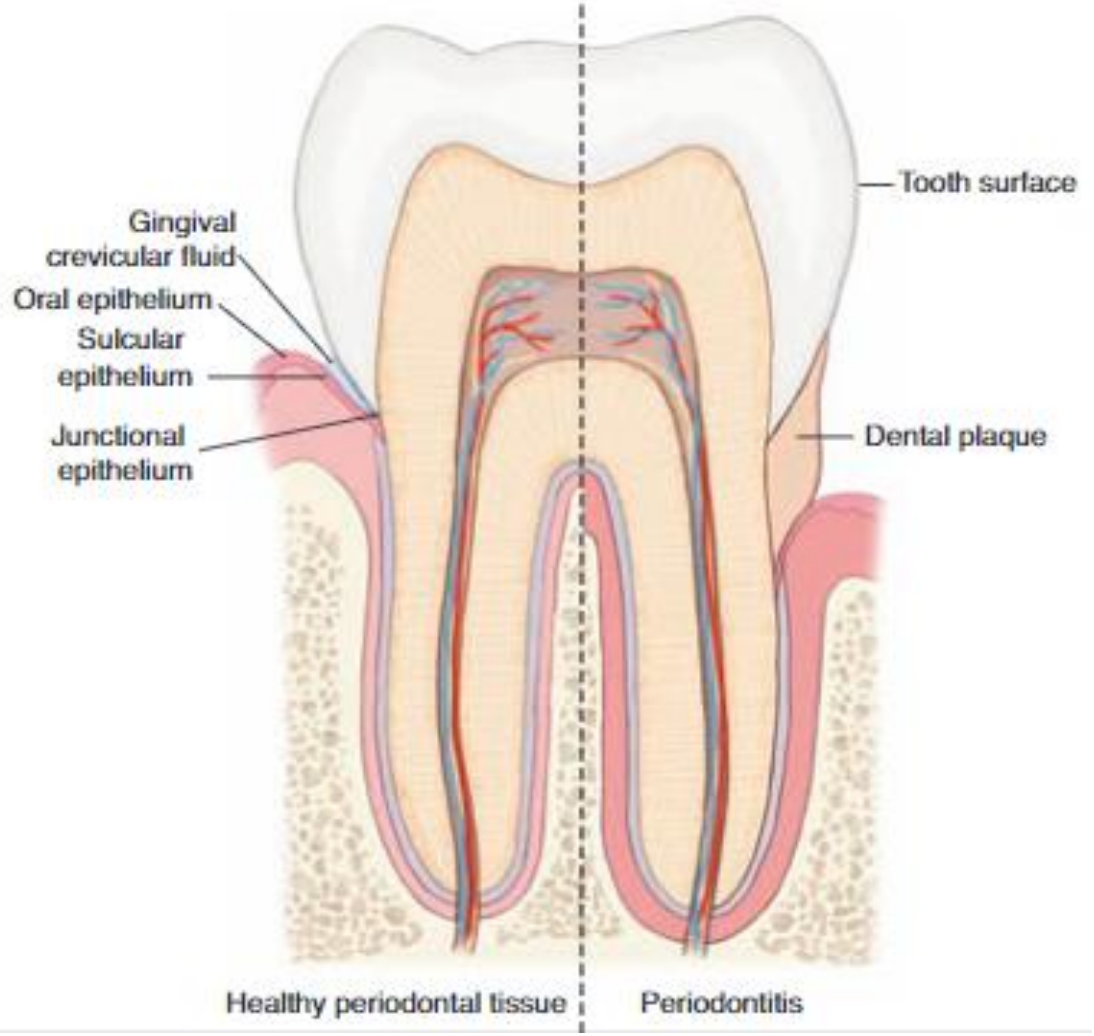
Connective tissue– supragingival fibers intact

PDL – attaches to root/bone

Bone – supports tooth



Normal Periodontium and Effects of Periodontal Disease



Healthy periodontal tissue (left) has connective tissue and alveolar bone. Oral epithelium covers this tissue, and junctional epithelium connects it to the tooth. The region between epithelium and tooth surface is sulcus, filled with **gingival crevicular fluid**.

In periodontitis (right), tooth and root surfaces are covered with dental plaque, which with inflammatory response, causes destruction of periodontal tissues and bone.

Periodontal Health: is characterized by predominant inhabitants of oral cavity, mostly Gram-positive facultative species such as *Streptococcus sanguis, Streptococcus mitis, Streptococcus salivarius, Actinomyces viscosus, Actinomyces naeslundii,* and a few beneficial Gram-negative species such as *Veillonella parvula* and *Capnocytophaga ochracea*.

Exposure to periodontal pathogens

Courtesy of Dr. Wu

Loss of epithelial integrity in pockets

Bacterial penetration

Recurrent Transient Bacteremia

BIOFILM

↓

Bacteremia

↓

Inflammatory Response

Biofilms release and can lead to a variety of biologically active inflammatory products, including:

- Bacterial endotoxins
- Protein toxins
- Destructive enzymes
- C-reactive protein

- These products can cause gingival inflammation and breakdown and can enter the bloodstream

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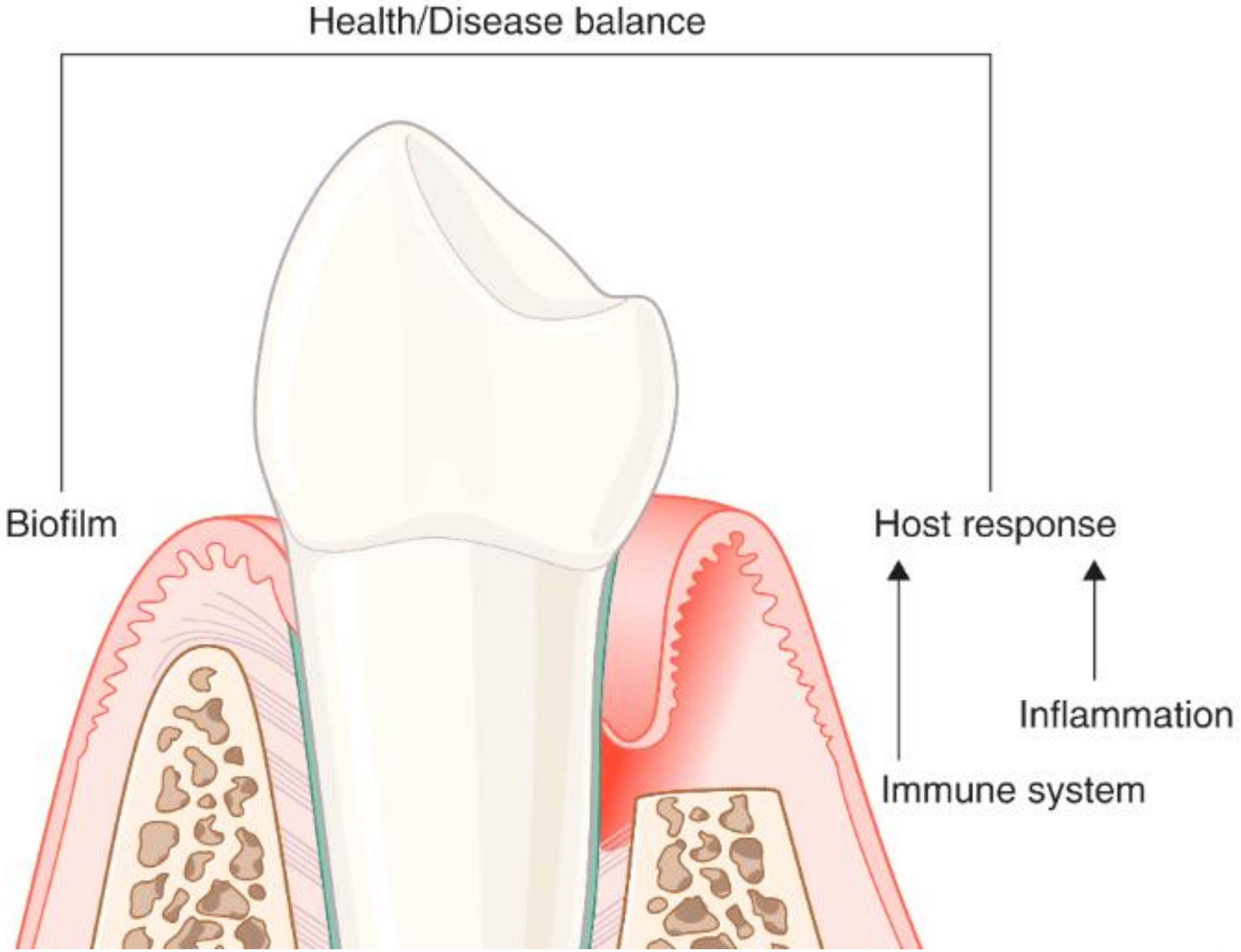
-Periodontal bacteria live in biofilm above and below the gingival margin
-Up to **1 billion** bacteria can inhabit a periodontal pocket
-Pocket depths typically range **4–12 mm**
-Biofilm alone does not cause disease—**host response + risk factors** (genes, environment, immune status) drive progression



Courtesy of Dr. Wu

Composition of Stage 1: Gingivitis

- Microbial insult leads to activation of leukocytes and stimulation of endothelial cells.
- Vascular changes lead to increased blood flow, widening of small capillaries.
- Leukocytes, polymorphonuclear neutrophils (PMN) leave capillaries and migrate to junctional epithelium and gingival sulcus.
- Time: 2-4 days after plaque accumulation
- Clinical findings: Potential bleeding on probing, plaque buildup
- Treatment: Ongoing monitoring, recall

Stage 2: Early Lesion

Courtesy of Dr. Wu

- Time: 4-7 days after plaque accumulation
- Blood vessels show vascular proliferation
- Predominant Immune Cells: are Lymphocytes
- Clinical findings: Bleeding on probing

- Microbial biofilm leads to local inflammation.
- Studies show more than 82 % of adolescents in US have gingivitis and bleeding.

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Courtesy of Dr. Wu

Stage 3: Established Lesion

- Time (days): 14-21 days after plaque accumulation
- Vascular proliferation of blood vessels
- Predominant immune cells are plasma cells and B Lymphocytes
- Clinical findings: Changes in color, texture, size
- In connective tissue, collagen fibers are destroyed.

Stage IV: Gingivitis to Periodontal Disease, Advanced Lesion

Courtesy of Dr. Wu

- Lesion extends into alveolar bone and periodontal breakdown.
- Gingivitis progresses to periodontitis in patients that may have multifactorial risk factors.

Histological changes

- Ulcerated epithelium
- Presence of inflammatory mediators and connective tissue breakdown
- Matrix metalloproteinases (MMPs) are widespread
- Repair of ulcerated epithelium depends on the regenerative activity of epithelial cells.

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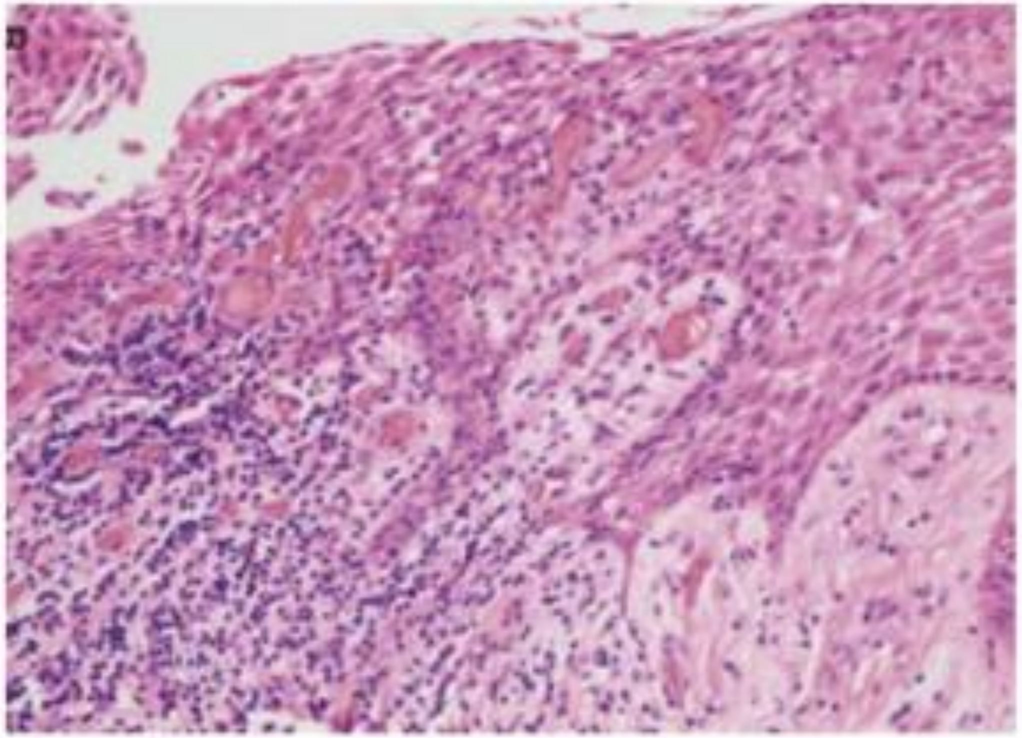
Courtesy of Dr. Wu

Color: Pale/coral pink	Erythematous
Consistency: Firm, resilient	Soft, edematous
Contour: Scalloped	Bulbous
Texture: Stippled	Smooth, shiny



RC, Schroeder HE: Lab Invest 33:235-249, 1976

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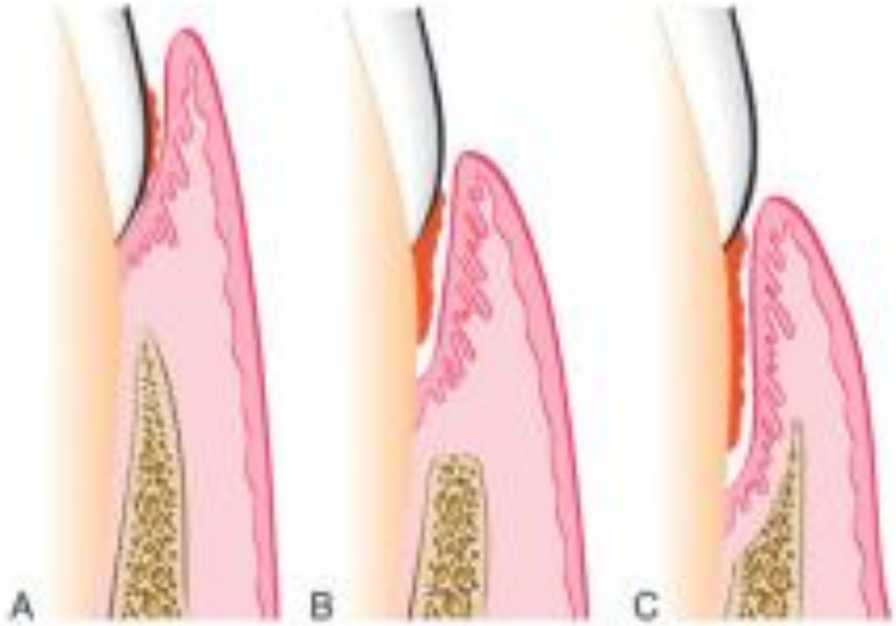
- Gingival pocket – gingival enlargement without bone loss
- Periodontal pocket – destruction of supporting bone and periodontal tissues



3 categories for periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis Associated with Systemic Diseases
- Periodontitis

Necrotizing Periodontal Disease

Clinical features

- Necrosis of papilla
- Bleeding
- Pain

Associated with impaired host Immune response.

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Periodontal Diseases

-Microbial Shift in Periodontal Disease: Shift toward Gram-negative, motile, anaerobic bacteria

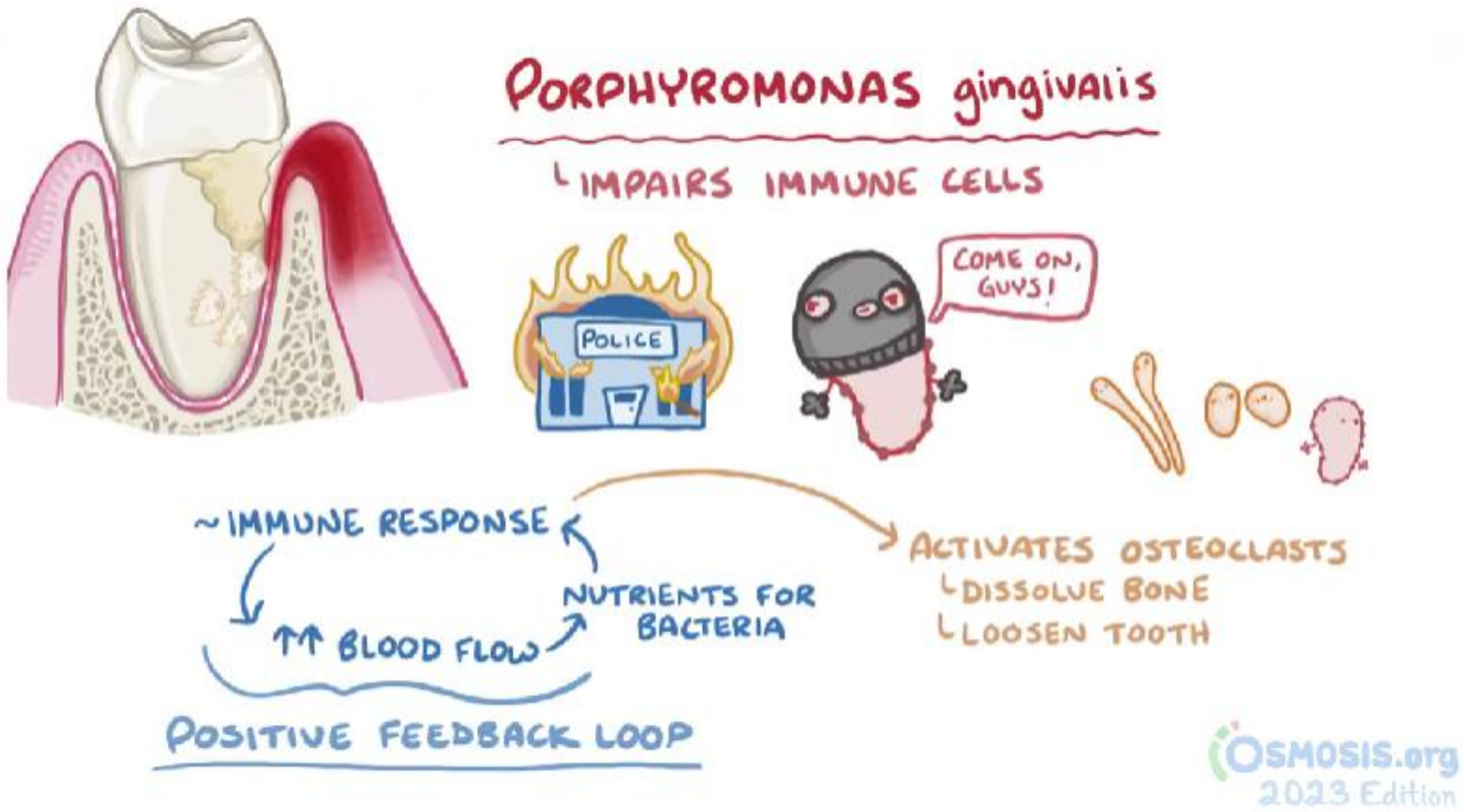
-Plaque: the primary etiologic factor

-Periodontal disease hypotheses:

1. **Non-specific plaque hypothesis:** periodontal disease results from elaboration of noxious products by the entire plaque flora. When only small amounts of plaque present, noxious products are neutralized by host. Large amounts of plaque would produce higher noxious products and overwhelm host's defense.
2. **Specific plaque hypothesis:** underlines the importance of qualitative composition of microbiota. Pathogenicity of dental plaque depends on presence/increase in specific microorganisms.
3. **Ecologic plaque hypothesis:** both the total amount of dental plaque and specific microbial composition of plaque may contribute to transition from health to disease.
4. **Keystone pathogen hypothesis and polymicrobial synergy and dysbiosis model:** certain low-abundance pathogens can orchestrate inflammatory disease by triggering the disruption of microbial homeostasis.

Figure 30-11 Normal gingiva with incipient gingivitis in tooth #7. Normal surface features are better revealed by drying gingiva.

Figure 30-12 Periodontal pockets around mandibular anterior teeth, showing rolled margins, edematous inflammatory changes, and abundant calculus and plaque.


PORPHYROMONAS gingivalis
IMPAIRS IMMUNE CELLS
POLICE
COME ON, GUYS!
~ IMMUNE RESPONSE
↑↑ BLOOD FLOW
NUTRIENTS FOR BACTERIA
POSITIVE FEEDBACK LOOP
ACTIVATES OSTEOCLASTS
DISSOLVE BONE
LOOSEN TOOTH
OSMOSIS.org
2023 Edition

Virulence Factors of Periodontopathogens

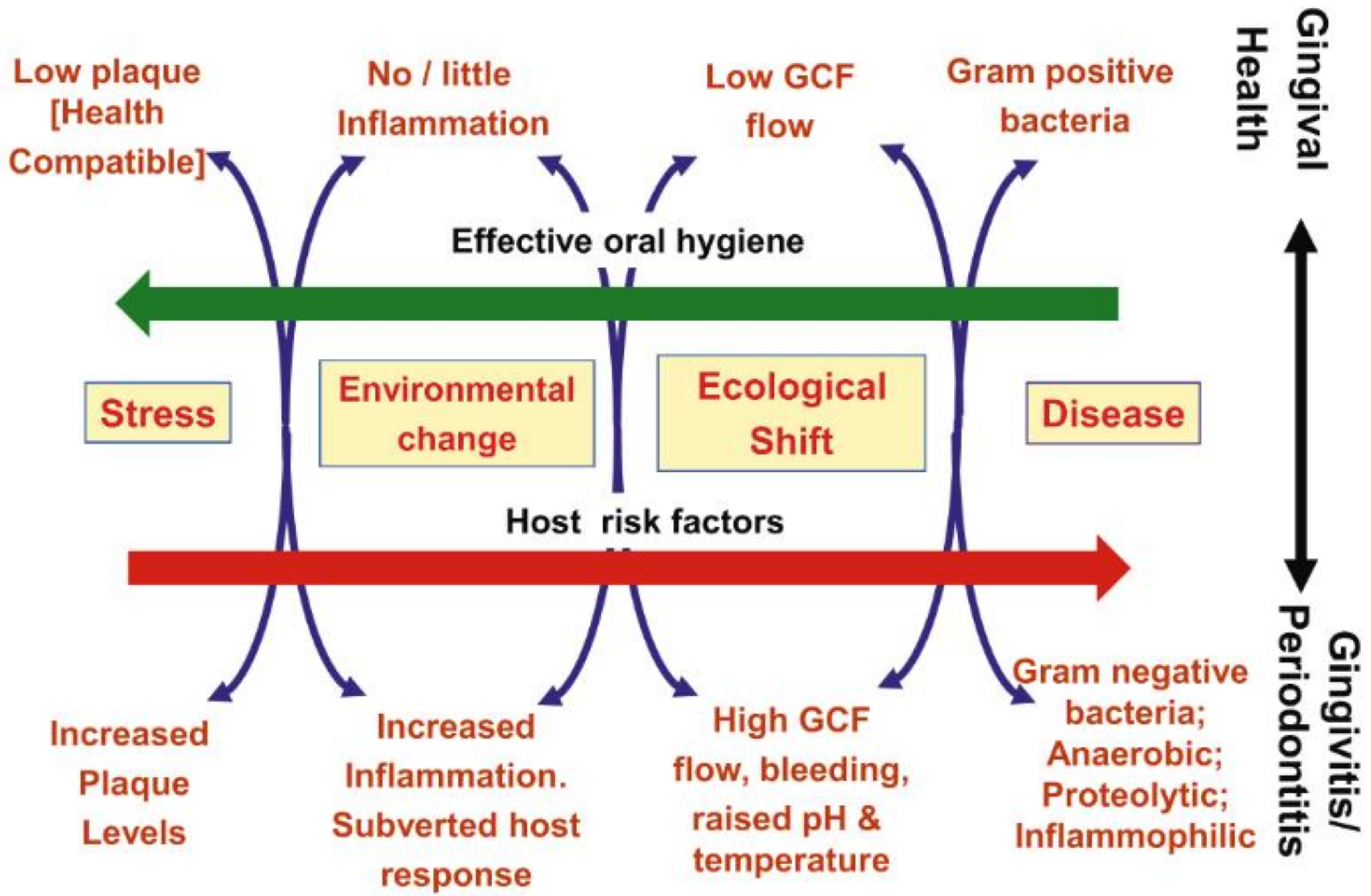
-Adhesion: Surface proteins and fibrils enable attachment to tissues and biofilm

-Tissue Destruction: Bacterial enzymes and proteins damage host tissues and trigger immune response/inflammation

-Immune Evasion: Strategies such as extracellular capsules

Microbial Shift: As plaque moves from **supragingival** to **subgingival** areas, flora shift from mainly **Gram-positive facultative cocci** to **Gram-negative anaerobic bacilli** and **spirochetes**

Disease Transition: Periodontal disease arises from interaction among a susceptible host, pathogenic bacteria, and reduced protective commensal species. Host susceptibility is influenced by genetics, environment, and behaviors (e.g., smoking, stress)



Mission: To advance the science and art of health through education, service, scholarship and social accountability

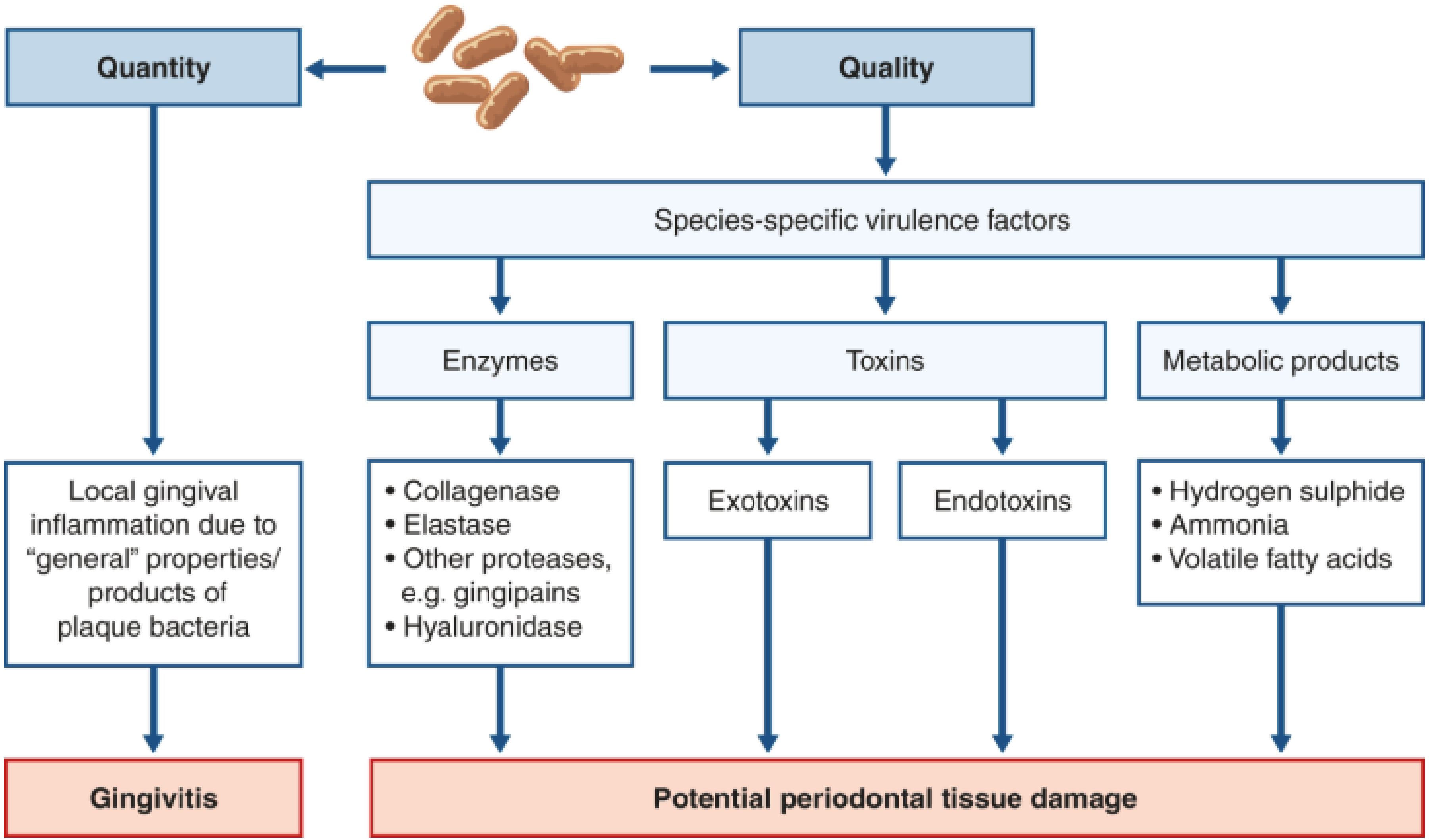
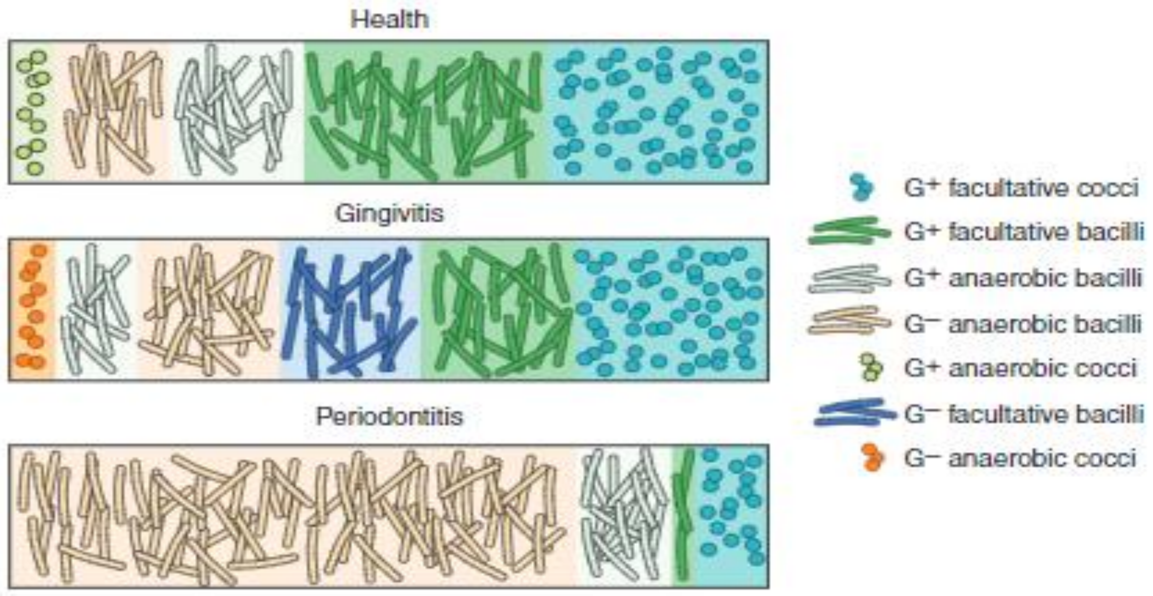


FIGURE 1.2.3 Overview of periodontitis pathogenesis.

Predominant plaque bacterial morphotypes in health, gingivitis, periodontitis



-Microbial Shifts: Health → Gingivitis → Periodontitis

Gram-positive → Gram-negative

Cocci → Rods → Spirochetes

Nonmotile → Motile

Facultative anaerobes → Obligate anaerobes

Sugar-fermenting → Proteolytic species

-Why the Shift Happens:

Early colonizers (Strep, Actinomyces) use oxygen → lower redox potential

Reduced oxygen favors anaerobes

Early bacteria use sugars; mature plaque supports species that use amino acids/peptides

"Chronic" Periodontitis Flora

Table is for illustrative purposes

- *Prevotella*
- *Bacteroides*
- *Porphrymonas gingivalis*
- *Eichenella*
- *Veillonella*
- *Capnocytophaga*
- *Actinobacillus*
- *Fusobacteria*
- spirochetes

TABLE 23-12 Summary of Bacterial Species Significantly Associated with Different Clinical Conditions (Based on Culture, Polymerase Chain Reaction (PCR), Checkerboard Data)

Health	Gingivitis	Chronic Periodontitis	Localized Aggressive Periodontitis	Aggressive Periodontitis	Necrotizing Periodontal Diseases	Abscesses of the Periodontium	Periimplantitis
Actinomyces gerencseriae *Actinomyces israelii* *Actinomyces naeslundii 1* *Actinomyces oris* *Actinomyces viscosus* *Capnocytophaga gingivalis* *Capnocytophaga ochracea* *Capnocytophaga* spp. *Capnocytophaga sputigena* *Eubacterium saburreum* *Fusobacterium periodonticum* *Fusobacterium polymorphum* *Neisseria* spp. *Propionibacterium acnes* *Streptococcus anginosus* *Streptococcus gordonii* *Streptococcus mitis* *Streptococcus oralis* *Streptococcus sanguinis* *Veillonella parvula*	*Actinomyces naeslundii 1* *Actinomyces viscosus* *Capnocytophaga* spp. *Streptococcus anginosus* *Streptococcus mitis* *Streptococcus oralis* *Streptococcus sanguinis* *Veillonella parvula* *Aggregatibacter actinomycetemcomitans* serotype a *Campylobacter concicus* *Campylobacter rectus* *Eikenella corrodens* *Eubacterium nodatum* *Fusobacterium nucleatum* *Haemophilus* spp. *Parvimonas micra* *Prevotella intermedia* spirochetes *Streptococcus intermedius* *Treponema* spp.	*Aggregatibacter actinomycetemcomitans* (serotype b) *Campylobacter rectus* *Eikenella corrodens* *Eubacterium nodatum* *Fusobacterium nucleatum* *Parvimonas micra* *Prevotella intermedia* Spirochetes *Treponema* spp. *Enterobacteriaceae* spp. *Leptotrichia buccalis* *Porphyromonas gingivalis* *Selenomonas noxia* *Tannerella forsythia* *Treponema denticola* *Prevotella nigrescens*	*Capnocytophaga* spp. *Aggregatibacter actinomycetemcomitans* high (Ltx clone) *Campylobacter rectus* *Eikenella corrodens* *Eubacterium nodatum* *Fusobacterium nucleatum* *Prevotella intermedia* Spirochetes *Treponema* spp. *Leptotrichia buccalis* *Porphyromonas gingivalis* *Tannerella forsythia* *Treponema denticola*	*Aggregatibacter actinomycetemcomitans* (serotype b) *Eubacterium nodatum* *Prevotella intermedia* *Spirochetes* *Treponema* spp. *Porphyromonas gingivalis* *Tannerella forsythia* *Treponema denticola* *Campylobacter gracialis* *Campylobacter showae* *Prevotella nigrescens*	*Fusobacterium nucleatum* *Prevotella intermedia* Spirochetes *Treponema* spp.	*Fusobacterium nucleatum* *Parvimonas micra* *Prevotella intermedia* Spirochetes *Porphyromonas gingivalis*	*Capnocytophaga* spp *Aggregatibacter actinomycetemcom* *Campylobacter rectus* *Fusobacterium nuclea* *Parvimonas micra* *Prevotella intermedia* Spirochetes *Treponema* spp *Enterobacteriaceae sp* *Porphyromonas gingiv* *Tannerella forsythia* *Treponema denticola* *Pseudomonas aerugin* *Staphylococcus aureu* *Staphylococcus epidermidis*

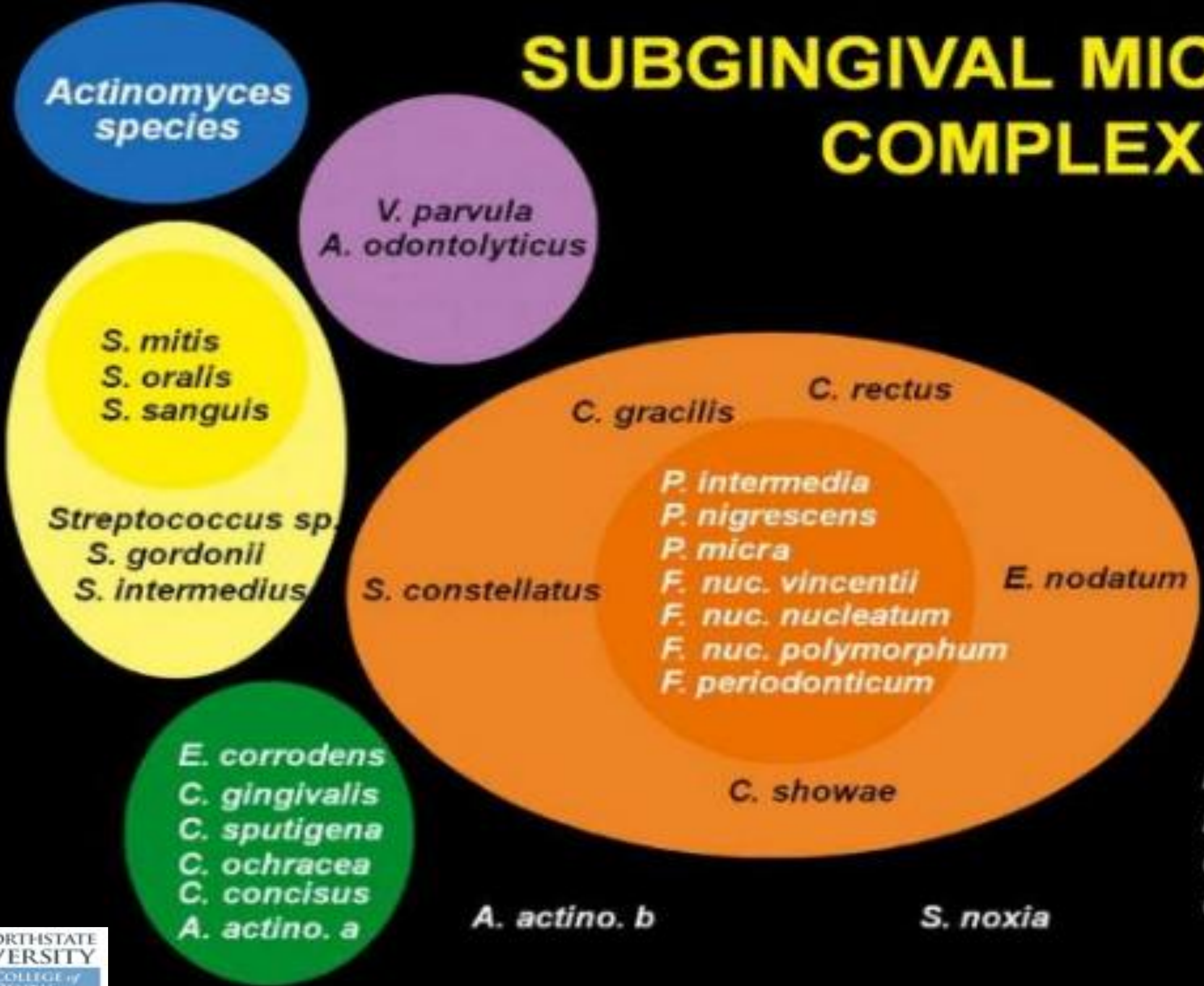
Ltx, Leukotoxin.

FIGURE 1.4.1 A patient with untreated chronic periodontitis.

FIGURE 3.1.5 Interproximal cratering following an episode of necrotising gingivitis

FIGURE 1.6.5 Generalised periodontitis in a heavy smoker.

SUBGINGIVAL MICROBIAL COMPLEXES



N subjects = 185
N samples = 13,261
Cluster analyses
Community ordination

California Northstate University
College of Dental Medicine

Courtesy of Dr. Goss lecture:

-The Socransky grouping of bacteria into complexes to reflect their interaction with host in health and periodontal disease. **Red complex:** found mostly in deep periodontal pockets, usually **preceded** by **bacteria** in **orange complex**. Members of **yellow**, **green**, and **purple complexes** usually found in **healthy** sites.

-Colored fonts refer to designation by **Socransky et al. Increased colonization** with **orange complex (*P intermedia*, *F nucleatum*, and *Peptostreptococcus micros*),** leads to more sites being colonized by **red complex**.

-Red complex: ***P. gingivalis***, ***T. forsythia***, and ***T. denticola***, found mostly in deep periodontal pockets. Sites containing **none** of these have the shallowest pocket depth.

-Microorganisms in **healthy** gingival sulcus generally **Gram positive** and **facultatively anaerobic** organisms. In chronic **periodontitis**, 75% **Gram negative**, 90% of that **strict anaerobes** and **motile bacilli/spirochetes**. Pathogenic potential of periodontal microorganisms is determined by their ability to colonize the host, evade host defenses, and damage host tissues.

-**Key periodontopathogens:** ***Porphyromonas gingivalis***, ***Prevotella intermedia***, ***Tannerella forsythia***, ***Aggregatibacter actinomycetemcomitans*** (formerly ***Actinobacillus actinomycetemcomitans***), ***Fusobacterium nucleatum***, ***Capnocytophaga*** species, and ***Treponema denticola***.

Dental Caries: bacteria driven, site-specific, dynamic, chronic disease

Characteristics:

-Very slow progression

-Multifactorial

-Infectious and transmissible

Caries formation needs:

-Cariogenic bacteria

-Susceptible surface

-Fermentable carbohydrate source

Key Oral Bacteria in Dental Caries

-Lactic Acid Bacteria (Streptococcus and **Lactobacillus):**

Use lactic acid fermentation (convert glucose → lactate)

Aciduric (tolerate acid) and **acidogenic** (produce acid)

-Streptococci are the primary acid producers in the mouth

-Lactobacillus species (e.g., *L. casei*) are present in low numbers but increase in caries as secondary invaders

-Streptococcus Mutans:

Primary initiator of dental caries

Adheres strongly to tooth surfaces via extracellular capsule

Produces **glucosyltransferase**, forming glucans that enhance adhesion

Generates **lactic acid**, lowering pH and causing enamel demineralization

Transmitted mainly from caregivers; infants acquire it only after tooth eruption

Caries Classification

- Pit and fissure
- Smooth-surface
- Root
- Recurrent

FIGURE 1.6.10 Root caries, subgingival restoration margins and calculus acting as local risk factors for periodontitis.

-Streptococcus *sp.* (*S mutans*, *S sanguis*, *S salivarius*) generally cause pit and fissure, smooth-surface, and root caries. *Streptococcus mutans* is the primary etiologic agent of caries.

-Lactobacillus sp. (*L casei*, *L acidophilus*) generally cause pit and fissure caries

-Actinomyces sp. (*A viscosus*, *A naeslundii*) generally cause root caries

Bacterial Plaque as Agent of Caries

-Oral Microorganisms and Disease:

Cause caries, periodontitis, endodontic infections, alveolar osteitis, tonsillitis

Linked to systemic diseases: CVD, diabetes, stroke, preterm birth, pneumonia

Many infections (caries, periodontitis, otitis media) arise from biofilm organisms

-Key Bacteria in Dental Plaque:

Streptococci (most abundant oral bacteria)

Gram-positive cocci; aerobic → facultative anaerobes

Major groups: mutans, salivarius, mitis, anginosus

S. mutans: aciduric, produces lactic acid → caries

S. sanguis: produces hydrogen peroxide

S. salivarius: found in saliva and soft tissues (tongue)

S. mitis: produces hydrogen peroxide

-Other Significant Species:

Lactobacilli (e.g., *L. casei*): highly acidogenic

Bifidobacterium spp.

Actinomyces spp.

Bacterial Plaque as Agent of Caries

***Predominant groups in subgingival plaque:** *Actinomyces, Prevotella, Porphyromonas, Fusobacterium, Veillonella*

***Streptococci have some essential properties necessary for caries formation:**
-Adhere and colonize to tooth surface
-Producing lactic acid dissolving enamel
-Producing a polymeric substance (from metabolism of carbohydrates), causing acid to remain in contact with tooth

-**Bacteria preferentially colonizing tooth surface:** *Streptococcus sanguis, Streptococcus mutans, Streptococcus mitis, Actinomyces viscosus*

-Streptococcus mutans: the primary etiologic agent **initiating dental caries**. The major cariogenic property of ***S mutans:*** its ability to produce the enzyme **glucosyltransferase** (dextran sucrase), giving the plaque a stickiness to adhere to tooth surfaces. Glucosyltransferases use sucrose as a substrate in synthesis of glucans (e.g. dextran, mutans, levans). Glucan production contributes to formation of dental plaque.

-S mutans produces 3 and ***S sobrinus*** produces 4 glucosyltransferases

-*S sanguis*, *S salivarius*, and *Lactobacillus* species produce glucosyltransferase

Hypotheses to Explain Initiation of Caries

1-**Specific plaque hypothesis:** emphasizes the importance of *mutans streptococci* in caries initiation.

2-**Non-specific plaque hypothesis:** heterogeneous groups of bacteria that can produce acids as end products of fermentation are involved in caries' initiation.

S mutans: the most effective Streptococcus species that causes caries, can initiate and maintain growth and continue acid production (primarily lactic acid) in sites with a low PH.

Lactobacillus species: present in increased numbers in most caries lesions on enamel and root surfaces and produce lactic acid (acidogenic), suggesting *lactobacilli* are involved in initiation of caries, but their affinity for tooth surface is low, their numbers in dental plaque in early caries lesions are low. Their numbers in saliva increase only after caries development. Thus, lactobacilli may **not** be involved in initiation of dental caries, but are involved in progression of caries deep into enamel and dentin (**pioneer organisms in advancing carious process**)

3-**Ecological plaque hypothesis:** Cariogenic flora found in natural plaque are weakly competitive and comprise only a minority of total community. Increase in fermentable carbohydrates results in prolonged low pH, promoting growth of acid-tolerant bacteria and initiating demineralization. Balance in plaque community turns in favor of *mutans streptococci* and *lactobacilli*. There is a dynamic relationship between bacteria and host, and changes in major host factors (e.g. salivary flow) can affect plaque development.

4-**Extended caries ecological hypothesis:** caries process consists of 3 reversible stages:

1. Intact enamel surfaces harbor mainly non-mutans streptococci and Actinomyces, causing only mild and infrequent acidification: stage of dynamic stability where demineralization and remineralization rates are in equilibrium or there is a net mineral gain.

2. More extensive and frequent acidification with a frequent supply of sugar. Non-mutans streptococci may adapt to these conditions by increasing their acidogenicity and acidurance. This shifts the balance between demineralization and remineralization toward net mineral loss and initiation of caries.

3. Acidic conditions are prolonged, more aciduric bacteria are selected, and more dominant microorganisms are mutans streptococci, lactobacilli, aciduric non-mutans streptococci, bifidobacteria, Actinomyces, and yeasts.

In the extended ecological plaque hypothesis, **not only mutans** streptococci but also the **entire population of acidogenic and aciduric bacteria** participate in the development of caries process. Thus, removal of specific aciduric bacteria, such as mutans streptococci, by antibiotics, vaccination, or gene therapy may **not** be sufficient for control of caries in the long term.

***Gram-positive cocci:**

-Oral **streptococci** are termed ***viridans streptococci*** because many produce green (viridis) colonies on blood agar. But many oral **streptococci** have strains that cause all 3 types of hemolysis.

-Mutans group (***mutans streptococci***) colonizes tooth, rapidly produces acid, and can grow in an acidic environment (**primary pathogen in enamel and root surface caries**). Antigen I/II proteins of ***S mutans*** are the surface components of bacterium that interact with pellicle.

-Mitis **group:** ***S sanguinis***, ***Streptococcus gordonii***, ***Streptococcus mitis***, and ***Streptococcus oralis***. These, except ***S mitis***, may become opportunistic and cause **infective endocarditis**. ***S sanguinis*** and ***S gordonii*** are **early colonizers** of tooth surface. ***S oralis*** and ***S mitis*** are among **the most common streptococci** in oral cavity and **initial colonizers**.

Other gram-positive cocci:** ***Granulicatella adiacens is an early colonizer of teeth. Anaerobic **cocci** such as ***Peptostreptococcus stomatis***, ***Parvimonas micra***, and ***Finegoldia magna*** often found in dental abscesses, carious dentin, infected pulp chambers and root canals, and advanced periodontal disease. ***Enterococcus faecalis*** has been isolated from infected root canals and periodontal pockets not responding to therapy.

Gram-positive bacilli and filaments:** ***Lactobacilli increases in advanced caries of enamel and root surface. ***Lactobacillus casei***, ***Lactobacillus fermentum***, ***Lactobacillus rhamnosus***, ***Lactobacillus salivarius***, ***Lactobacillus acidophilus***, and ***Lactobacillus gasseri*** are some of the most common (acidogenic and acid tolerant).

***Eubacteria:** pleomorphic, obligate anaerobes found in dental plaque and periodontal pockets.

***Actinomyces species:** (short **rods** or pleomorphic with branching) are major component of dental plaque.

Actinomyces israelii: may be filamentous, can be an opportunistic pathogen, and cause actinomycosis, characterized by chronic orofacial inflammation (could cause infection which creates a sinus tract from the apex of tooth through skin). Yellow deposits known as "sulfa granules" could be found at infection site).

Actinomyces naeslundii: the most common **Actinomyces** species in dental plaque, in root surface caries and gingivitis.

Actinomyces radicidentis: found in endodontic infections.

Actinomyces viscosus and ***naeslundii*** cause root surface caries.

***Gram-negative cocci:**

Neisseria species help in plaque formation by consuming oxygen and obligate anaerobes flourish.

Veillonella species: anaerobic gram-negative **cocci** found mostly in plaque (cannot metabolize carbohydrates and can metabolize lactic acid)

*Gram-negative bacilli:

Haemophilus parainfluenzae: facultatively anaerobic gram-negative bacillus

Aggregatibacter actinomycetemcomitans: found in subgingival plaque, associated with particularly aggressive forms of juvenile periodontitis and less frequently with adult periodontitis. Also associated with endocarditis, subcutaneous and brain abscesses, and osteomyelitis.

Capnocytophaga: carbon dioxide–dependent (capnophilic), fusiform rods with motility, found in subgingival plaque, associated with destructive periodontal disease. Include: ***Capnocytophaga gingivalis***, ***Capnocytophaga sputigena***, ***Capnocytophaga ochracae***, ***Capnocytophaga granulosa***, and ***Capnocytophaga haemolytica***

Eikenella corrodens: found in dental plaque, associated with dentoalveolar abscesses, infective endocarditis, some forms of chronic periodontitis

Fusobacterium species: strict anaerobes, slender “cigar” shape. ***Fusobacterium nucleatum*** isolated mainly from periodontal pockets. **Fusobacteria** thought to be an important bridging organism between early and late colonizers during plaque formation. Together with **spirochetes**, involved in inflammatory periodontal disease. “***fusospirochetal* complex**” cause necrotizing ulcerative gingivitis and necrotic oral ulcers.

Porphyromonas species: pleomorphic, asaccharolytic **bacilli**, utilizing proteins and peptides for metabolism, are strict anaerobes. Produce **black-pigmented** colonies on blood agar. ***Porphyromonas gingivalis*** is found almost exclusively at subgingival sites, esp. in advanced periodontal lesions. ***Porphyromonas endodontalis*** identified primarily in infected root canals.

Prevotella: pleomorphic, strict anaerobes. ***Prevotella intermedia***, ***Prevotella melaninogenica***, ***Prevotella nigrescens***, and ***Prevotella loescheii*** form **black-pigmented** colonies. ***P intermedia*** is associated with chronic periodontitis and endodontic infections.

-Gingivitis associated with hormone fluctuations (pregnancy, puberty, menstrual cycle, oral contraceptive use) is associated with elevation of ***Prevotella intermedia***, which uses steroids as growth factors.

Campylobacter rectus: curved bacillus with polar flagella, strict anaerobe, found in periodontal disease.

Actinobacillus species: gram-negative coccobacillary rods
Treponema: motile, helical cells (spirochaete), strict anaerobes
Mycoplasma: smallest free-growing cells, not have a peptidoglycan layer, may play a role in salivary gland hypofunction and periodontal disease

Box 24-1 The prevalent phyla, genera, and species found in root canal infections

Image is for illustrative purposes

Actinobacteria
- *Actinomyces species*
- *Propionibacterium acnes*
- *Propionibacterium propionicum*

Bacteroidetes
- *Tannerella forsythia*
- *Porphyromonas endodontalis*
- *Porphyromonas gingivalis*
- *Prevotella species*

Firmicutes
- *Enterococcus faecalis*
- *Eubacterium species*
- *Lactobacillus species*
- *Parvimonas micra*
- *Peptostreptococcus species*
- *Selenomonas species*
- *Streptococcus species*
- *Veillonella parvula*

Fusobacteria
- *Fusobacterium nucleatum*

Proteobacteria
- *Eikenella corrodens*
- *Campylobacter gracilis*
- *Campylobacter rectus*

Spirochetes
- *Treponema denticola*
- *Treponema maltophylum*
- *Treponema parvum*
- *Treponema socranskii*

Red, most abundant; blue, abundant; purple, less abundant. (Data from Lamont et al.)

Interesting Facts

-Oral microbes include: bacteria, fungi, protozoa, viruses (e.g., herpesviruses, HPV, picornaviruses, retroviruses)

-Age and **plaque:**

Older adults do not form plaque faster

Plaque levels may be higher due to **reduced dexterity**

-Individual variation:

Some people form plaque faster than others

Being a "heavy plaque former" **does not justify poor oral hygiene**

FIGURE 1.6.11 Significant supragingival calculus formation impeding adequate biofilm control.

Summery

-Periodontal health:

Mainly gram-positive facultative species (e.g., *Streptococcus* spp., *Actinomyces* spp.)
Some beneficial gram-negative species (*Veillonella parvula*, *Capnocytophaga ochracea*)

-Plaque-induced gingivitis:

Mix of gram-positive and gram-negative, facultative and anaerobic cocci, bacilli, spirochetes
Includes *Streptococcus*, *Actinomyces*, *Peptostreptococcus*, *Fusobacterium*, *Prevotella*, *Campylobacter*

-Chronic periodontitis:

Predominantly gram-negative anaerobic bacilli and spirochetes
Key species: *P. gingivalis*, *T. forsythia*, *T. denticola*, *Prevotella*, *Fusobacterium*, *Eikenella*, *Campylobacter*

-Microbial shift:

Healthy sulcus: mostly gram-positive facultative anaerobic bacteria
Periodontitis: ~75% gram-negative, ~90% of that strict anaerobes, motile bacilli and spirochetes

-Pathogenicity depends on:

Ability to colonize, evade host defenses, and damage tissues

Microorganism	Relevance
Streptococcus sp. (*S mutans, S sanguis, S salivarius*)	Pit and fissure, smooth-surface, and root caries
Lactobacillus sp. (*L casei, L acidophilus*)	Pit and fissure caries
Actinomyces sp. (*A viscosus, A naeslundii*)	Root caries
Streptococcus mutans	The primary etiologic agent of caries
P. gingivalis, T. forsythia, and *T. denticola* (found mostly in deep periodontal pockets)	The red complex
P intermedia, F nucleatum, and *Peptostreptococcus micros* (usually precedes the red complex)	The orange complex
Porphyromonas gingivalis, Prevotella intermedia, Tannerella forsythia, Aggregatibacter actinomycetemcomitans (formerly *Actinobacillus actinomycetemcomitans*), *Fusobacterium nucleatum, Capnocytophaga* species, and *Treponema denticola*	Key periodontopathogens

Microorganism	Relevance
***S mutans*, *S sobrinus*, *S Sanguis* (the most frequently isolated Streptococcus in oral cavity), *S Salivarius*, and *Lactobacillus* species**	**Produce glucosyltransferase**
Actinomyces*, *Prevotella*, *Porphyromonas*, *Fusobacterium*, and *Veillonella	**The predominant groups in subgingival plaque**
S Sanguis* and *S Mutans	**Predominant Streptococci found in dental plaque**
Streptococci	**The predominant supragingival bacteria**
***Prevotella intermedia* (uses steroids as growth factors)**	**Gingivitis associated with hormone fluctuations (pregnancy, puberty, menstrual cycle, oral contraceptive use)**

Resources

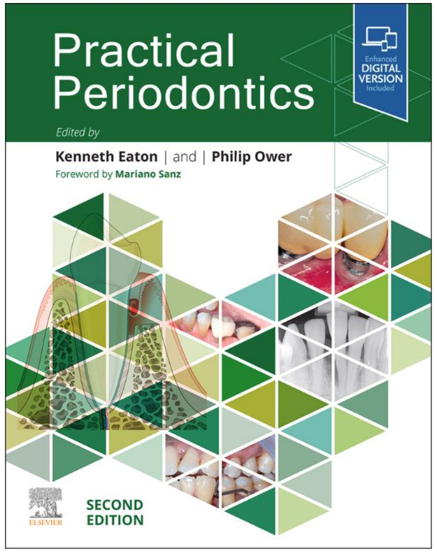
-Medical Microbiology and Immunology for Dentistry, Düzgüneş, Nejat, 2016

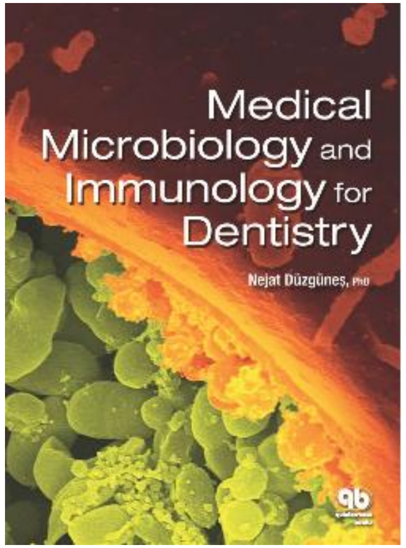
-Oral Microbiology and Immunology, 3rd Edition, George N. Hajishengallis, Richard J. Lamont, Hyun (Michel) Koo, Howard F. Jenkinson

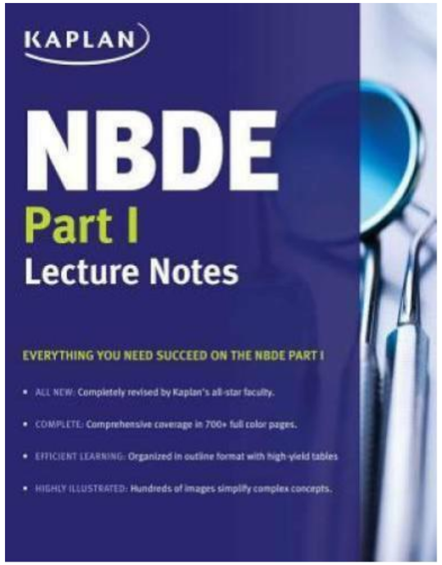
-Carranza's Clinical Periodontology, 13th Edition

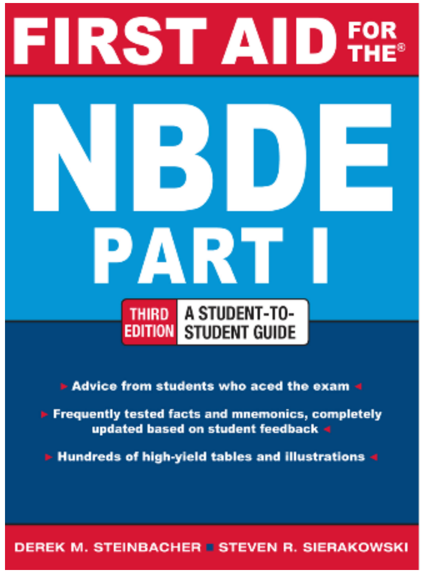
-First Aid NBDE Part I

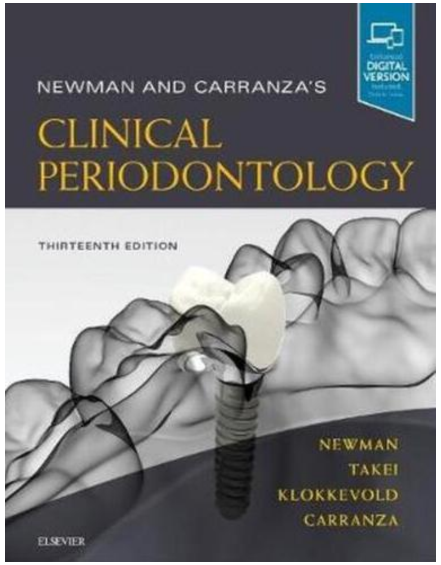
-Kaplan NBDE Part I

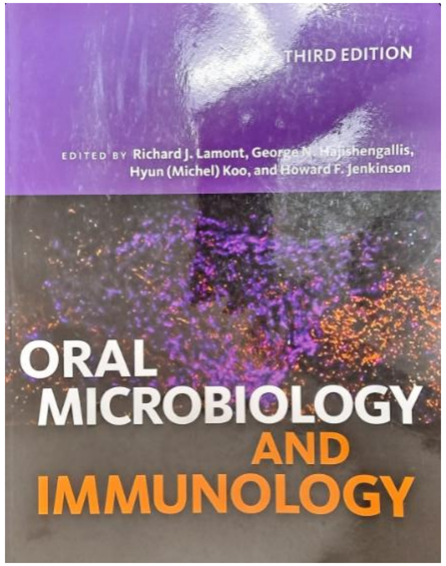












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